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SUBMITTED TO:

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Terminal Lab - Cypress E2E Testing

COURSE TITLE:

Software Automated Testing

COMSATS UNIVERSITY ISLAMABAD

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1. Task Overview

This report documents the completion of Part B of the terminal lab using the Cypress framework for end-to-end testing. The selected public web application was Sauce Demo, which provides login, shopping cart, checkout, and page navigation flows suitable for functional UI testing.

All required test cases from Task 1 and Task 2 were implemented, executed on the local PC, and verified with real Cypress test output and browser screenshots.

2. Cypress Project Setup

Item	Details
Website under test	https://www.saucedemo.com
Framework	Cypress 13.17.0
Browser used by run	Electron 118 (headless)
Node runtime	Node.js v20.11.1 local runtime
Run command	npm test / cypress run
Spec folder	cypress/e2e/task1 and cypress/e2e/task2

3. Task 1 - UI Test Suite (Functional Testing)

Task 1 required login, navigation, and form tests. Every implemented test used `cy.visit()`, `cy.get()`, and `cy.should()` as required.

Required Test	Spec File	Main Action	Assertion Result
Login Test 1	task1/login-valid.cy.js	Entered valid username and password.	Products page displayed and URL included /inventory.html.
Login Test 2	task1/login-invalid-password.cy.js	Entered valid username with incorrect password.	Error message was visible.
Login Test 3	task1/login-empty-validation.cy.js	Submitted empty login fields.	Username required validation appeared.
Navigation Test 1	task1/navigation.cy.js	Clicked shopping cart link after login.	Cart page title and URL were verified.
Navigation Test 2	task1/navigation.cy.js	Visited the app, logged in, then moved from inventory to cart.	Products and Your Cart headings were verified in sequence.
Form Test 1	task1/form.cy.js	Filled checkout first name, last name, and postal code.	Checkout overview and summary information were visible.

4. Task 2 - Assertions, Aliases & Custom Commands

Task 2 extended the suite with assertion practice, a negative assertion, alias usage, a reusable custom command, and beforeEach hooks.

Exercise	Implementation	Evidence
Assertion Practice	Used <code>should('be.visible')</code> , <code>should('have.text', ...)</code> , and <code>should('have.attr', ...)</code> on different login page elements.	task2/assertions_aliases.cy.js passed.

Negative Assertion	Verified that the login error element does not exist before submission.	Negative assertion passed.
Alias Practice	Saved the username input with <code>as('usernameInput')</code> and reused it with <code>get('@usernameInput')</code> .	Alias test passed.
Custom Command	Created <code>cy.login()</code> in <code>cypress/support/commands.js</code> and used it in a test.	Custom command test passed and screenshot captured.
beforeEach Hook	Used <code>beforeEach()</code> to call <code>cy.visit()</code> before each Task 2 test.	Hook executed across Task 2 tests.

5. Test Execution Result

The complete Cypress suite was executed locally on the PC using the command below:

npm test

Specs	Tests	Passing	Failing
7	12	12	0

Final Cypress result: All specs passed in 00:16 with 12 passing tests, 0 failing tests, 0 pending tests, and 0 skipped tests.

6. Difficulty Faced and Solution

One difficulty was choosing stable selectors that would not break when the page layout changed. I solved this by using Sauce Demo data-test attributes wherever possible because they are intended for automated testing. Another challenge was keeping Task 1 and Task 2 separate while still avoiding repeated login code. I kept Task 1 explicit because the instructions required core Cypress commands in each test, and used a custom command only in Task 2. Sauce Demo also does not support direct deep-link visits to the cart route, so the navigation test opens the app and reaches the cart through the actual UI flow.

7. Appendix - Real Cypress Screenshots

The following screenshots were captured by Cypress during the real local test execution.

Screenshot 1: Checkout form test reached the overview page

Swag Labs

Checkout: Overview

QTY	Description
1	Sauce Labs Backpack carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection. \$29.99

Payment Information:
SauceCard #31337

Shipping Information:
Free Pony Express Delivery!

Price Total
Item total: \$29.99
Tax: \$2.40
Total: \$32.39

Cancel

Finish



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Screenshot 2: Valid login landed on Products page

Swag Labs

ProductsName (A to Z)



Sauce Labs Backpack

carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection.

\$29.99

Add to cart



Sauce Labs Bike Light

A red light isn't the desired state in testing but it sure helps when riding your bike at night. Water-resistant with 3 lighting modes, 1 AAA battery included.

\$9.99

Add to cart



Sauce Labs Bolt T-Shirt

Get your testing superhero on with the Sauce Labs bolt T-shirt. From American Apparel, 100% ringspun combed cotton, heather gray with red bolt.

\$15.99

Add to cart



Sauce Labs Fleece Jacket

It's not every day that you come across a midweight quarter-zip fleece jacket capable of handling everything from a relaxing day outdoors to a busy day at the office.

\$49.99

Add to cart



Sauce Labs Onesie

Rib snap infant onesie for the junior automation engineer in development. Reinforced 3-snap bottom closure, two-needle hemmed sleeved and bottom won't unravel.

\$7.99

Add to cart



Test.allTheThings() T-Shirt (Red)

This classic Sauce Labs t-shirt is perfect to wear when cozying up to your keyboard to automate a few tests. Super-soft and comfy ringspun combed cotton.

\$15.99

Add to cart

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Screenshot 3: Navigation test opened the cart page



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Your Cart

QTY

Description

← Continue Shopping

Checkout



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Screenshot 4: Navigation sequence reached the cart page



Swag Labs



Your Cart

QTY

Description

← Continue Shopping

Checkout



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Screenshot 5: Custom `cy.login()` command reached Products page



Swag Labs



Products

▼ Name (A to Z) ▼



Sauce Labs Backpack

carry.allTheThings() with the sleek, streamlined Sly Pack that melds uncompromising style with unequaled laptop and tablet protection.

\$29.99

Add to cart



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